

On the Transport Mechanics

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Abstract

In these brief notes a basic description of quantum transport through a chaotic cavity, such as the array of QDs proposed by Bordin et. al [1] will be introduced, in order to justify further simulation of the conductance of that model.

SNS Junctions scattering

We remind that our system is made of a chain of quantum dots coupled by superconducting leads; each of the QDs is gated and the superconductors are grounded.

We are interested in the middle junction, as in Fig.1, which is a SNS junction, following Beenakker's [2] theory of transport, the excitation spectrum of quasiparticles is given by the solutions to the following BdG equation:

$$\begin{pmatrix} H_0 & \Delta \\ \Delta^* & -H_0 \end{pmatrix} \begin{pmatrix} \Psi_e \\ \Psi_h \end{pmatrix} = E \begin{pmatrix} \Psi_e \\ \Psi_h \end{pmatrix} \quad (1)$$

Where,

$$\Psi_e = \begin{pmatrix} \Psi_e^+ \\ \Psi_e^- \end{pmatrix}; \quad \Psi_h = \begin{pmatrix} \Psi_h^+ \\ \Psi_h^- \end{pmatrix} \quad (2)$$

And $\Delta = \Delta(x)1 \otimes i\sigma_y$, since it's the only term which couples electron-hole. On the other hand, H_0 is a 2x2 matrix which has the on-site energies, and the electron-electron hopping strengths. We will assume no spin involved since it's a Kitaev model. We will treat the system as a Josephson junction in a black box formalism, which is universal for these kind of phenomena [3], as in Fig. 2.

We will now make an important assumption: since the InSb/Al hybrids are short, we will assume to be in the short junction limit, namely $L \ll \xi$, for ξ being the BCS coherence length. We will summarize schematically the following thought process:

1. In the S region, ideally $\Delta(\vec{r}) = \Delta_0 \exp(\mp i\frac{\phi}{2})$ in the limit $|x| \rightarrow \infty$;
2. Near the junctions, Δ is subject to proximity effects: it has to satisfy boundary conditions ;
3. The proximity effects are relevant in a small volume around the junctions, out of which normal superconductivity is restored;
4. In the short junction limit, since $\xi \gg L$, cooper pairs are unaffected by proximity effects, and the pairing Δ follows a step function:

$$\Delta(x) = \begin{cases} \Delta_0 \exp(+i\frac{\phi}{2}) & x < -\frac{L}{2} & \text{superconductive region} \\ 0 & |x| < \frac{L}{2} & \text{normal region} \\ \Delta_0 \exp(-i\frac{\phi}{2}) & x > \frac{L}{2} & \text{superconductive region} \end{cases} \quad (3)$$

We can now treat the different scattering processes separately, meaning that at the S/N junction there will be Andreev reflections, while in the N region a chaotic description of scattering will be needed. Since Andreev reflection arise at the junctions, it mean that below the superconductive excitation spectra the energy levels will be populated by Andreev bound states (ABS). These are finite in number since there will only be a finite number of conductive modes allowed. In the normal disordered region, on the other hand, there will be scattering processes that won't couple electron-holes, meaning the scattering matrix will be block diagonal. We will fix a position $|x_0| < \frac{L}{2}$ in the normal region and study the scattering processes locally.

$$\hat{S}(x_0) = \begin{pmatrix} \hat{S}_e & 0 \\ 0 & \hat{S}_h \end{pmatrix} \quad (4)$$

According to the electron-hole symmetry, we require $\hat{S}_e = \hat{S}$ and $\hat{S}_h = \hat{S}^\dagger$, Where:

$$\hat{S} = \begin{pmatrix} r_{LL} & t_{LR} \\ t_{RL} & r_{RR} \end{pmatrix} \quad (5)$$

Because of conservation laws, $\hat{S}$ is unitary, that is $\hat{S}^\dagger \hat{S} = 1$. Furthermore, by definition, $\Psi^{out} = \hat{S} \Psi^{in}$.

We now switch our interest to the scattering processes in the outer disordered region, that is:

$$\begin{cases} -\frac{L}{2} < x < x_0 \\ x_0 < x < \frac{L}{2} \end{cases} \quad (6)$$

The starting point is to put ourselves in the BdG basis. Since we are in a conductive normal-metal medium, and the conduction takes place along the x-axis, the system will be anisotropic: along the x axis we will assume ballistic transport, i.e. the length of the system L is small compared to the mean free path l and the solutions will be that of free wave. On the transverse plane a potential will confine the electron motion and the solution won't be trivial. The bulk hamiltonian will thus be:

$$H_0 = \left(\frac{p_y^2 + p_z^2}{2m} \right) + V(y, z) + \frac{P_x^2}{2m} - E_F \quad (7)$$

With electron-hole eigenfunctions in the N_1 lead:

$$\begin{cases} \Psi_{n,e}^\pm(N_1) = \begin{pmatrix} 1 \\ 0 \end{pmatrix} (K_n^e)^{-\frac{1}{2}} \Phi_n(y, z) e^{\pm i K_n^e (x + \frac{L}{2})} \\ \Psi_{n,h}^\pm(N_1) = \begin{pmatrix} 0 \\ 1 \end{pmatrix} (K_n^h)^{-\frac{1}{2}} \Phi_n(y, z) e^{\pm i K_n^h (x + \frac{L}{2})} \end{cases} \quad (8)$$

Where $K_n^{e,h} = (\frac{2m}{\hbar^2})^{\frac{1}{2}} (E_F - E_n + \sigma^{e,h} \epsilon)^{\frac{1}{2}}$, E_n is the eigenvalue of the transverse mode n, $\Psi_n(y, z)$, and $\sigma^{e,h} = \pm 1$, respectively. For the lead N_2 the solution are the same, but translated in $(x - \frac{L}{2})$.

We could find an analogous set of eigenfunctions in the superconducting outer region. Before accounting for boundary conditions, we would introduce a naming convention: We label the incoming/outgoing modes if they are incoming/outgoing the normal disordered region at x_0 , as shown in Fig.2. The subscript N indicates N conducting modes.

$$\begin{cases} c_N^{in} = (c_e^\dagger(N_1), c_e^-(N_2), c_h^-(N_1), c_h^\dagger(N_2)) \\ c_N^{out} = (c_e^-(N_1), c_e^\dagger(N_2), c_h^\dagger(N_1), c_h^-(N_2)) \end{cases} \quad (9)$$

Since the normal region scattering does not couple e-h, $c_N^{out} = \hat{S}_N c_N^{in}$, where $\hat{S}_N$ is of the block diagonal form discussed above. The electron-hole symmetry implies $\hat{S}_h(E) = \hat{C} \hat{S}_e(-E)^\dagger \hat{C}^{-1}$, where $\hat{C}$ is unitary. Each of the block diagonal elements will then satisfy $S^{ee}(\epsilon) = S^{hh}(-\epsilon)^*$.

The reflection and transmission matrices, r_{ij} and t_{ij} , are $N \times N$ since there are N conducting modes.

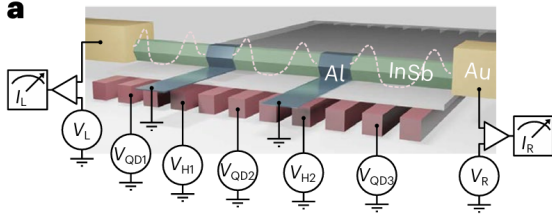


Figure 1: Experimental setup. A InSb nanowire is places on an array of eleven gates and contacted by two Al(blue) and two Cr/Au (yellow) leads.

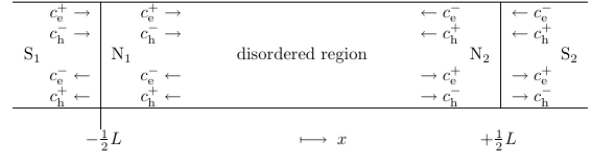


Figure 2: Black box treatment of a superconductor-normal metal-superconductor Josephson junction containing a disordered normal region. Scattering states in the normal and superconducting leads are indicated schematically.

We will assume the same number of modes for electron and holes, meaning each of the scattering matrices will be in total $2N \times 2N$.

We will now switch our attention to the energies at which various scattering processes can occur: we have already established that Andreev reflection will occur for energies $0 < \epsilon < \Delta$ at the N/S junction. The related scattering matrix will now couple e-h, but, since it's a reflection, from the disordered region point of view will only have the off-diagonal reflection terms:

$$S_A = \alpha \begin{pmatrix} 0 & r_A \\ r_A^* & 0 \end{pmatrix}; \quad r_A = \begin{pmatrix} e^{i\frac{\theta}{2}} \mathbf{1} & 0 \\ 0 & e^{-i\frac{\theta}{2}} \mathbf{1} \end{pmatrix} \quad (10)$$

We have $C_N^{in} = \hat{S}_A C_N^{out}$. We assume the number of modes to still be N because, since the energy range is below the superconductive gap Δ , we can neglect additional normal transport mechanism, that would otherwise change the number of modes. That is the so called Andreev approximation. The last step is to focus on the energy region $\epsilon > \Delta$, but, since the Andreev reflections do not occur in this region, it is justified to write the scattering matrix for the whole SNS junction. We will now label the incoming and outgoing states with the subscript S , since we will treat the superconductive regions and the leads and are only interested in the scattering in/out these regions.

$$\begin{cases} c_S^{in} = (c_e^\dagger(S_1), c_e^-(S_2), c_h^-(S_1), c_h^\dagger(S_2)) \\ c_S^{out} = (c_e^-(S_1), c_e^\dagger(S_2), c_h^\dagger(S_1), c_h^-(S_2)) \end{cases} \quad (11)$$

Again $c_S^{out} = S_{SNS} c_S^{in}$. Calling for boundary conditions we get:

$$S_{SNS} = U^{-1}(1 - M)^{-1}(1 + M^\dagger)S_N U \quad (12)$$

Where:

$$U = \begin{pmatrix} r_A & 0 \\ 0 & r_A^* \end{pmatrix}^{\frac{1}{2}}; \quad M = \alpha S_N \begin{pmatrix} 0 & r_A \\ r_A^* & 0 \end{pmatrix} \quad (13)$$

Before treating all of the above scattering matrices, we should ponder over the short junction ansatz we made in the beginning. The whole energy spectrum is illustrated in Fig. 3 In the short junction limit, namely $L \ll \xi$, that is $\Delta_0 \ll E_c = \frac{\hbar \bar{v}}{\tau_{dwell}}$, it can be shown [3] that the density of states related to the continuum spectrum does not contribute to the Josephson current $I(\phi)$, that is $\frac{\partial \rho}{\partial \phi} = 0$. We will then only consider discrete energy states. From the N region point of view:

$$\begin{cases} c^{in} = \hat{S}_A c^{out} \\ c^{out} = \hat{S}_N c^{in} \end{cases} \quad (14)$$

The total scattering matrix is just the product (they form a unitary group) of the scattering matrices for each process, meaning $c^{in} = \hat{S}_A \hat{S}_N c^{in}$. The problem is now to just solve:

$$\det(\mathbf{1} - \hat{S}_A \hat{S}_N) = 0 \quad (15)$$

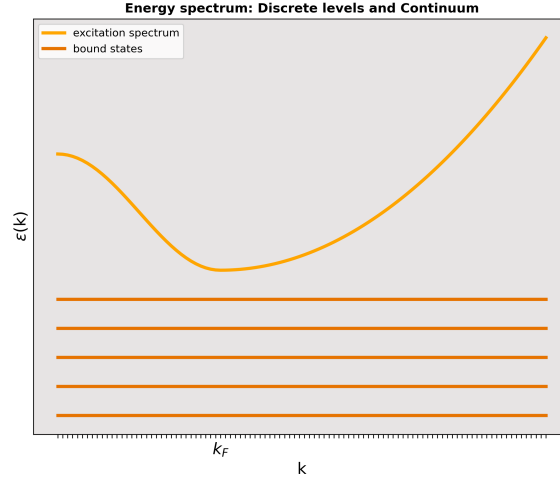


Figure 3: Toy model for the total energy spectrum: the top curve is the continuum BCS excitation spectrum, while the lower lines are a depiction of the bound, discrete states below the pairing Δ .

Random matrix Theory and Weidenmuller's formula

We will briefly introduce basic concepts of the random matrix theory [4] of quantum transport, following [2]. Quantum transport in a chaotic cavity (Fig.4) can occur via many different scattering processes, meaning there is an ensemble of Hamiltonians that are able to evolve the system. Following canonical statistical mechanics, Wigner and Dyson introduced the following ensemble: $P(H) = e^{-\beta \text{Tr} V(H)}$, where $V(H)$ is a suitable potential. In the case $V(H) \propto H^2$, the ensemble is said to be gaussian. The ensemble space is of dimension N , where each of these matrices is $M \times M$ dimensional. For them to be hermitian, we require the parameters to be either real, complex, or real quaternions¹. We can then define the parameter $\beta = 1, 2, 4$ for real, complex and real quaternion elements respectively, representing their degree of freedom, and it renormalizes the pdf of the ensemble. Since the hamiltonians are hermitian, we can diagonalize them, as such: $H = UDU^{-1}$, where D is the diagonal eigenvalues matrix, and U is the eigenvector matrix. Since the trace is cyclic, that is $\text{Tr}(H) = \text{Tr}(UDU^{-1}) = \text{Tr}(UU^{-1}D) = \text{Tr}(D)$, then $\text{Tr}V(H) = \sum_n V(E_n)$, meaning $P(H)$ does not depend on the U matrices. It is then implied that the U matrices are distributed uniformly over the ensemble space. The only meaningful distribution is the one of the eigenvalues $\{E_n\}$, which are distributed according to a measure: $d_\mu H = d\mu U \prod_n dE_n J$, where J is the jacobian. It can be shown that if the energy levels are discrete, the distribution assumes a Gibbs like form: $P(E_n) = c \exp(-\beta(\sum_{i < j} \ln|E_i - E_j| + \sum_i V(E_i)))$.

The logarithmic term is interpreted as a Coulomb repulsion between energy terms. Microscopic parameters play a role in the potential V , but do not correlate the energy levels E_n with each other. If that was the case, we would deviate from a logarithmic repulsion. The correlations between the energy levels are to be attributed to geometrical reasons, such as the shape and dimension of the system, which render the whole system chaotic, or non-integrable. The only necessary condition to apply this theory is that the only constants have to be the energy levels.

In order to apply this theory in the quantum dot case, some hypothesis have to be made first:

1. The material is a good metal, meaning $E_{c,closed} \gg \delta$, where δ is the mean level energy spacing.
2. The classical motion is chaotic, happening at timescales larger than the ergodic timescale $\tau_{ergodic}$. The hamiltonian of such a system is compatible with the Wigner and Dyson ensemble, on energy scales lower than the Thouless energy, $E_{c,closed} \simeq \frac{\hbar}{\tau_{ergodic}}$.
3. It does not matter whether the transport is ballistic or diffusive.

¹Real quaternions are 2x2 matrices obtained as $q = \alpha \mathbf{1} + i\beta \sigma_x + i\gamma \sigma_y + i\delta \sigma_z$. Said matrices are symplectic if they are unitary with real valued quaternions elements.

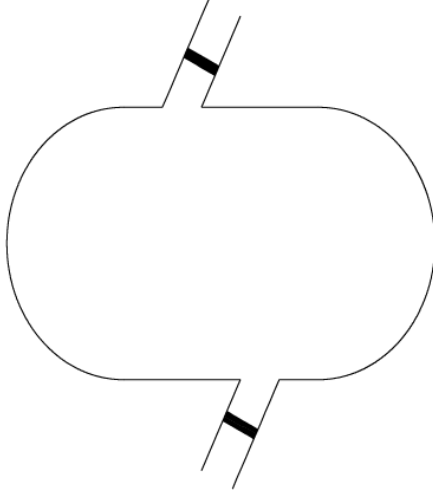


Figure 4: Depiction of a chaotic cavity system, which is open via two leads. There is a strong dependence on initial conditions to the electronic transport, meaning different scattering processes will occur, and the scattering matrices S will be distributed accordingly.

4. In a good metal we have $\lambda_F \ll l$, with l being the mean free path. This means that the wave functions are extended and not localized.

Now, by coupling the system to the leads (see Fig.4), it will transition from a closed system to an open one, since we are coupling it to reservoirs brought out of equilibrium. We can thus make a further hypothesis: if we are in the weak coupling regime, meaning $\tau_{dwell} > \tau_{ergodic}$, and the open system Thouless energy $\frac{\hbar}{\tau_{dwell}} \simeq E_{c,open} \ll E_{c,closed}$, the system keeps its chaotic nature. It follows that $\delta \ll E_{c,closed}$.

Under these hypothesis the transport is independent of microscopic properties such as the shape of the cavity. The only property that matters is if the ensemble is coupled to ballistic point contacts, in which case the ensemble will be gaussian, or diffusive, for which the ensemble will be poissonian. The next step is to focus on the scattering matrix approach.

There are N_1 conductive modes in the first lead, and N_2 in the second one. For ballistic leads, the number of modes is the only meaningful property. In general $N_1 \neq N_2$. We would now introduce a more computationally friendly approach to the gaussian ensemble, the circular ensemble:

$$P(\phi_n) \propto \prod_{n < m} |e^{i\phi_n} - e^{i\phi_m}|^\beta \quad (16)$$

Which is the most general ensemble which describes phase relationships in a chaotic cavity. By definition ², S is distributed uniformly over the unitary group. Since we are interested in the transport mechanics, we would like to focus on the transport matrix eigenvalues $\{T_n\}$, and their statistical distribution. Generally, the transmission matrices will be $N_1 \times N_2$ dimensional, but we will restrict to the case where the number of modes in the two leads are the same, and the matrices will just be squared $N \times N$ matrices.

But what does it mean that the cavity is chaotic?

Chaotic processes have to maximize entropy, defined as:

$$\mathbf{S} = - \int d\mu(S) P(S) \ln P(S) \quad (17)$$

$\mathbf{S}$ is subject to different constraints as follows: for $\beta = 1$ it is symmetric; for $\beta = 4$ is self dual; an additional constraint called the analyticity-ergodicity constraint: analytic because S has poles only on

²In the circular ensemble the S matrix distribution is uniform over the unitary sphere manifold representation of the unitary group.

the lower half of the complex energy plane, meaning that, since the system is open, charge carriers can dwell towards the leads in a characteristic time τ_{dwell} , and the energies will take an imaginary part:

$$E_n \rightarrow E_n - \frac{i\Gamma_n}{2} \quad (18)$$

Which is responsible of a Lorentzian broadening of the energy levels; It is then ergodic because in a chaotic cavity the whole ensemble of state will be explored, in such a way that the ensemble average corresponds to the spectral energy average, that is $\int d\mu(S) S^P P(S) = \bar{S}^P \quad \forall P \in \mathbf{N}$. Under these hypothesis, all of the pdf momenta will be determined by the average, sub unitary³ matrix $\bar{S}$. The ensemble that maximizes the entropy and respects such constraints is the poissonian one:

$$P(S) \propto |\det(\mathbf{1} - \bar{S}^\dagger S)|^{-\beta(N_1+N_2-1+\frac{2}{\beta})} \quad (19)$$

The circular ensemble is a special case of the poissonian one, for which $\bar{S}^\dagger = 0$ and $P(S) = \text{const.}$. The poissonian ensemble is best suited for tunnel leads, while the circular for ballistic ones. we will now go back to the black box description of transport through a SNS junction.

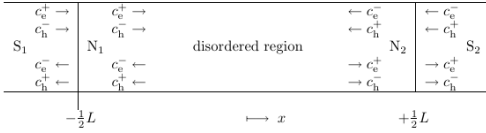


Figure 5: Black box depiction of an SNS junction. In each sequence a different scattering process, described by its scattering matrix S , will occur.

If the step-like pairing condition introduced in the beginning is met (meaning a short normal section in between the leads), we can split the various scattering processes, and, since the S matrices form a unitary group, the total scattering matrix will be just the product: $S = S_0 S_1 S_2$. It is distributed according to the Poisson ensemble. The average scattering matrix will only have $N \times N$ reflection terms at the leads:

$$\bar{S} = \begin{pmatrix} r_1 & 0 \\ 0 & r_2 \end{pmatrix} \quad (20)$$

The eigenvalues of $(\mathbf{1} - \bar{S}\bar{S}^\dagger)$ are the tunnel probabilities Γ_n through the leads. It can be shown [2] that the scattering approach is microscopically equivalent to the hamiltonian approach. The most generic hamiltonian of a chaotic cavity couples to tunnel leads is:

$$H = \sum_a |a\rangle E_F \langle a| + \sum_{\mu,\nu} |\mu\rangle H_{\mu\nu} \langle \nu| + \sum_{\mu,a} (|\mu\rangle W_{\mu a} \langle a| + h.c.) \quad (21)$$

Where $a = 1, \dots, N$, $N = N_1 + N_2$ is the total number of modes via the leads, and $\mu = 1, \dots, M$ is the number of bound states in the cavity. The matrix terms:

1. $H_{\mu\nu}$ is a $M \times M$ hermitian matrix, with real ($\beta = 1$), complex, ($\beta = 2$) or real quaternion ($\beta = 4$) elements.
2. $W_{\mu a}$ are coupling terms which make up a $M \times N$ matrix W .

Such a hamiltonian is distributed by a gaussian ensemble:

$$P(H) \propto \exp(-\beta(\frac{\pi}{2\delta})^2 M^{-1} \text{Tr} H^2) \quad (22)$$

Where δ is the mean level spacing at the Fermi level for $M \rightarrow +\infty$. The average scattering matrix will then be:

³Subunitary meaning $\det(\bar{S}\bar{S}^\dagger) \leq 1$

$$\bar{S} = \frac{M\delta - \pi^2 W^\dagger W}{M\delta + \pi^2 W^\dagger W} \quad (23)$$

Since the eigenvalues of $(\mathbf{1} - \bar{S}^\dagger \bar{S})$ are Γ_n , we can relate Γ_n with the eigenvalues of W , ω_n :

$$\begin{cases} \Gamma_n = \frac{4M\delta\pi^2\omega_n}{(M\delta + \pi^2\omega_n)^2} \\ \omega_n = \frac{M\delta}{\pi^2\Gamma_n}(2 - \Gamma_n \pm 2\sqrt{1 - \Gamma_n}) \end{cases} \quad (24)$$

We are now ready to introduce the Weidenmuller formula for the scattering matrix, valid under those assumptions:

$$S = \mathbf{1} - 2\pi i W^\dagger (E_F - H + i\pi W W^\dagger)^{-1} W \quad (25)$$

Conductance

We will treat the interface as a perfect NS junction as in Fig. 6. The same assumptions of rigid boundary conditions discussed before will be made. Furthermore, another proximity effect will be discussed: other than the suppression of the pair potential Δ at the superconducting side of the NS interface, which is a negligible effect anyways, we have to consider the induction of a non-zero order parameter Ψ at the normal side of the NS interface. The order parameter quantifies the degree of phase coherence between electron and holes, but it is absent in the BdG equations, and will not contribute to the dynamics.

Following the discussion on the scattering matrix for Andreev reflections, we can explicitly derive the scattering matrix terms by implying boundary conditions at the NS interface. The idea is that Andreev reflections transform an electron mode into a hole mode, via a phase shift. Schematically:

$$\begin{cases} c_e^-(N_2) = \alpha e^{i\phi} c_h^-(N_2) \\ c_h^+(N_2) = \alpha e^{-i\phi} c_e^+(N_2) \end{cases} \quad (26)$$

Where $\alpha = e^{-i \arccos(\epsilon/\Delta_0)}$. Combining eq. 26 and eq. 11, we get the following:

$$\begin{cases} c_e^-(N_1) = S_{ee} c_e^+(N_1) + S_{eh} c_h^-(N_1) \\ c_h^+(N_1) = S_{he} c_e^+(N_1) + S_{hh} c_h^-(N_1) \end{cases} \quad (27)$$

The four $N \times N$ matrices $S_{ee}, S_{eh}, S_{he}, S_{hh}$ form together the scattering matrix S of the whole system for energies $0 < \epsilon < \Delta_0$. These can be interpreted as follows: an electron incident in lead N_1 is reflected either as an electron (with amplitudes S_{ee}), or as a hole (with amplitudes S_{he}). The other two matrices are the scattering amplitudes of hole-hole or hole-electron processes. The four matrices can be found according to the Weidenmuller formula 25.

The only quantity left to derive is the conductance. We will do so following [5]. It is assumed that the conductance takes place along the z-axis, as in Fig. 6. The total current in the disordered region will then be:

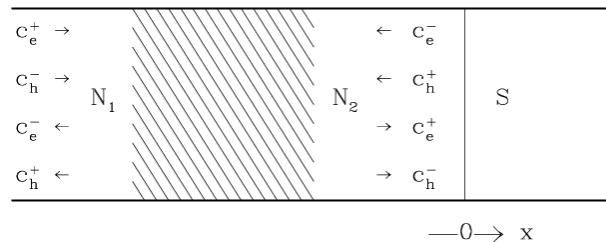


Figure 6: Normal-metal-superconductor junction containing a disordered normal region (shaded).

$$j_i(r) = \int d^3r' \sigma_{ij}(r, r') E_j(r') \quad (28)$$

Where $\sigma_{ij}(r, r')$ is the non local conductivity tensor. Since conduction only takes place on the z-axis it follows that the zero temperature conductance will be:

$$G_{mn}^{(0)} = \int_{c_m} d\rho \int_{c_n} d\rho' \sigma_{zz}(r, r') \quad (29)$$

The integration is performed over the cross sections m, n respectively. In order to derive the conductance coefficients, it is necessary to get the microscopic expression of the conductivity tensor, defined as[6]:

$$\sigma_{zz}(\omega) \simeq \int_0^\infty dt e^{i\omega t} \langle [j_z(t), j_z(0)] \rangle \quad (30)$$

For an oscillating driving field at frequency ω . In quantum mechanics, a magnetic field will split fermion energy levels and therefore break time reversal symmetry. For any such process the conductivity tensor will take an imaginary part. According to the fluctuation-dissipation theorem, the real and imaginary parts abide to the Kramers-Kronig relations. Following Kubo's work [6] we will split the conductivity tensor in its symmetric and antisymmetric parts, which will be complex in general. Two independent relations will be taken into account, namely:

$$\begin{cases} \sigma_{ij}(\omega) = \sigma'_{ij}(\omega) + i\sigma''_{ij}(\omega) \\ \sigma_{ij}(\omega) = \sigma^a_{ij}(\omega) + \sigma^s_{ij}(\omega) \end{cases} \quad (31)$$

In our case an external static magnetic field is applied to suppress spin, meaning TRS is broken at $\omega = 0$, and the tensor follows the static Onsager relation $\sigma_{ji}(-H) = \epsilon_i \epsilon_j \sigma_{ij}(H)$, where $\epsilon_k = \pm 1$ whether k is symmetric (antisymmetric) with respect to time reversal. We will then be concerned with the symmetric/antisymmetric components of the static conductivity tensor:

$$\begin{cases} \sigma_{\mu\nu}^s(0) = \frac{\beta}{2} \int_{-\infty}^\infty \Psi_{\mu\nu}^s(t) dt \\ \sigma_{\mu\nu}^a(0) = 2 \int_0^\infty \Psi_{\mu\nu}^a(t) \int_0^t \Gamma(\tau) d\tau dt \end{cases} \quad (32)$$

where $\Psi_{\mu\nu}^{a/s}(t) = Tr \rho \{j_\nu(0) j_\mu(T)\}$ is the correlation function of current components. We adopted the BdG formalism in the Nambu basis, for which the wave functions $\Psi^{s/a}$ are scattered via S. In deriving the scattered wave functions, one should adopt the Green functions (retarded/advanced) formalism as shown in [5]. The convolutionary kernel $\Gamma(t)$ is defined as:

$$\Gamma(t) = \frac{2}{\hbar\pi} \log \coth\left(\frac{\pi}{2\beta\hbar}\right) |t| \quad (33)$$

Which is clearly zero at zero temperature. By substituting 32 into 28 one gets the conductance at zero temperature:

$$G_{ij}^{(0)}(\omega) = G_0(\delta_{ij} - |S_{ij}^{ee}(\omega)|^2 + |S_{ij}^{he}(\omega)|^2) \quad (34)$$

Where G_0 is the conductance quantum $\frac{e^2}{h}$ (a factor of two would be needed due to spin degeneracy, but in our Kitaev model spin is suppressed). We now need to take into account temperature effects, meaning an antisymmetric part will arise. The assumption of ideal (perfectly transparent) NS interface is made. The conductance is then averaged over the energy spectra:

$$G_{ij}^{(T)}(\omega) = \int_{-\infty}^{+\infty} dE \frac{G_{ij}^{(0)}(E)}{4K_b T \cosh^2\left(\frac{E-\omega}{K_b T}\right)} \quad (35)$$

Under those assumptions, it follows directly from the Landauer formula.

Application to the poor man's Majorana

Fig. 1 depicts an open system when coupled to the Au leads. The microscopic model has to be described by a hamiltonian such as 21, containing on site energies and coupling terms between the quantum dots and the leads. Moreover, it has a Kitaev-like microscopic hamiltonian which couples the QDs to each other:

$$H = \sum_{n=1}^3 \mu_n c_n^\dagger c_n + \sum_{n=1}^2 (t_n c_n^\dagger c_{n+1} + \Delta_n c_n^\dagger c_{n+1}^\dagger + h.c.) \quad (36)$$

For which spin is suppressed by applying a magnetic field of the order of $\simeq 200mT$, which would account for a Zeeman energy splitting greater than the thermal fluctuation of the system at low enough temperatures. A representation of the energy landscape is presented in Fig. 7.

It can be noted immediately that there are at most 2 conduction modes for each quantum dot; furthermore, they have to realize resonant tunneling conditions for conductance to take place. The relevant parameters of the systems are: the left and right energies of the leads, μ_L, μ_R ; the on site energies μ_i of each quantum dot; the hopping terms t_n ; the superconducting pairing terms Δ_n ; the dot-lead coupling strengths Γ_L, Γ_R , for the left and right lead respectively; a finite temperature T .

Once the conditions for conductance are met, two main scattering processes can occur in our system: crossed Andreev reflections (CAR), which are basically Andreev reflections happening into neighboring quantum dots (the difference with an Andreev reflection is that there is not an NS junction, but a tunnel barrier between the quantum dots); elastic co-tunneling (ECT), namely electron tunneling between neighboring QDs. The resulting scattering matrix is of the type discussed in eq. 20 and eq. 15. Such processes are illustrated in Fig.8.

In the above-mentioned hypothesis the hamiltonian is described by a gaussian ensemble and a random matrix theory approach is valid. It should be noted that every parameter here is real, hence $\beta = 1$.

We will first treat the case where one of the quantum dots at the edge is kept off-resonance (see Fig. 9-10) , and the conduction happens between the first two quantum dots: the leftmost one is coupled to the left Au lead, while the rightmost one is grounded and is not coupled to any lead. We will make use of the Weidenmuller formula 25 for the scattering matrix, where $W = \text{diag}(\sqrt{\Gamma_L}, 0, -\sqrt{\Gamma_L}, 0)$ is the dot-lead coupling matrix, which is diagonal in this case, regarding a 2 site chain and particle-hole symmetry. Since one of the quantum dots is not coupled, its coupling term is 0. The matrix is 4×4 because we are in the Nambu basis, that is the hamiltonian can be written as

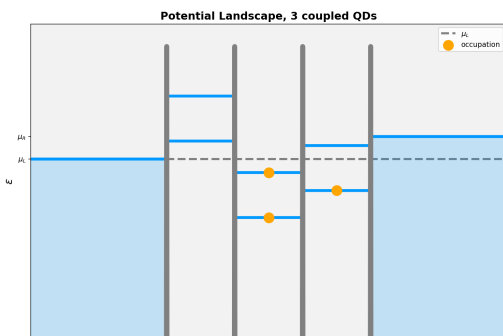


Figure 7: Potential landscape of the whole system: (from left to right) the on site chemical potential μ_L of the left lead; the three quantum dots energy level separated by tunnel barriers; the chemical potential of the right lead μ_R . At most 2 conduction modes are available per quantum dots, which have to be on resonance with each other.

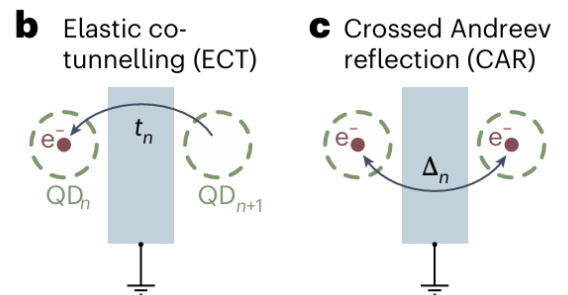


Figure 8: **b.** Elastic co-tunnelling (ECT) between neighboring quantum dots; **c.** Crossed Andreev reflection (CAR), creation or annihilation of two electrons into neighboring QDs.

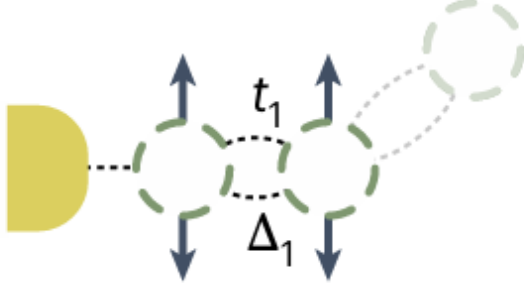


Figure 9: Schematic depiction of the system, when the third quantum dot is kept off resonance. Only the first QD is coupled to the lead, while the other is grounded.

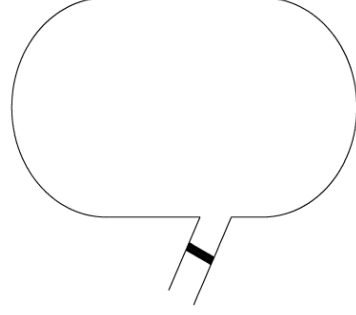


Figure 10: When one the QDs is grounded, the lead is closed. The system is then open only via the input lead, and the total scattering matrix will only have reflection terms.

$$\begin{cases} H = \frac{1}{2}\Psi^\dagger h \Psi \\ \Psi = (c_1, c_2, c_1^\dagger, c_2^\dagger)^T \end{cases} \quad (37)$$

We will then calculate the conductance as discussed above, factoring in the finite temperature convolution. The resulting plots are the so called charge stability diagrams. We will plot the diagram for the left lead differential conductance, $G_{LL} \equiv \frac{dI_L}{dV_L}$.

The following parameters have been assumed: since we are interested in resonant tunneling conditions, only the energy differences (see Fig.7) are relevant, as such we can just center everything at $(\mu_1, \mu_2) = (0, 0)\mu eV$ and, in order to achieve resonance also the chemical potential of the lead has to be sufficiently close to $0\mu eV$, namely $\mu_L \simeq 0.001\mu eV$. The thermal energy is taken to be $K_b T = 3\mu eV$, 0.3 times the coupling strengths $t_1 = \Delta_1 = 10\mu eV$. Accordingly, the dot-lead coupling strength is chosen to be $\Gamma_L = 1.5\mu eV$. Such parameters should theoretically realize the so called poor man's Majorana sweet spot condition, in which the PMM should be stable to perturbations of the parameters.

For the time being, we will focus on discussing the numerical result of the theoretical PMM sweet spot, as well as an interesting phenomenon: by off-setting the hopping strength and superconducting pairing term, one of the two main scattering processes (crossed Andreev reflections and elastic co-tunneling) will dominate, resulting in a phase transition in the diagram and the subsequent dislodgement of one or more crossings in the charge stability diagram (see Fig.11).

In theory, since we have 3 quantum dots, the occupation number that characterizes the system is a three-dimensional vector $|n_1, n_2, n_3\rangle$, where n_i represents the occupation number of the i -th quantum dot. By keeping the third dot off-resonance, we are only concerned with the two-dimensional sliced vector $|n_1, n_2, 0\rangle$, fixing the third occupation number to zero. Hence the plot is two dimensional. Furthermore, since the occupation number for each QD can only be $\{0, 1, 2\}$, the plot is subdivided into 9 different regions. Non zero conductance is observed on the transition from one occupation number to another, where resonance is met.

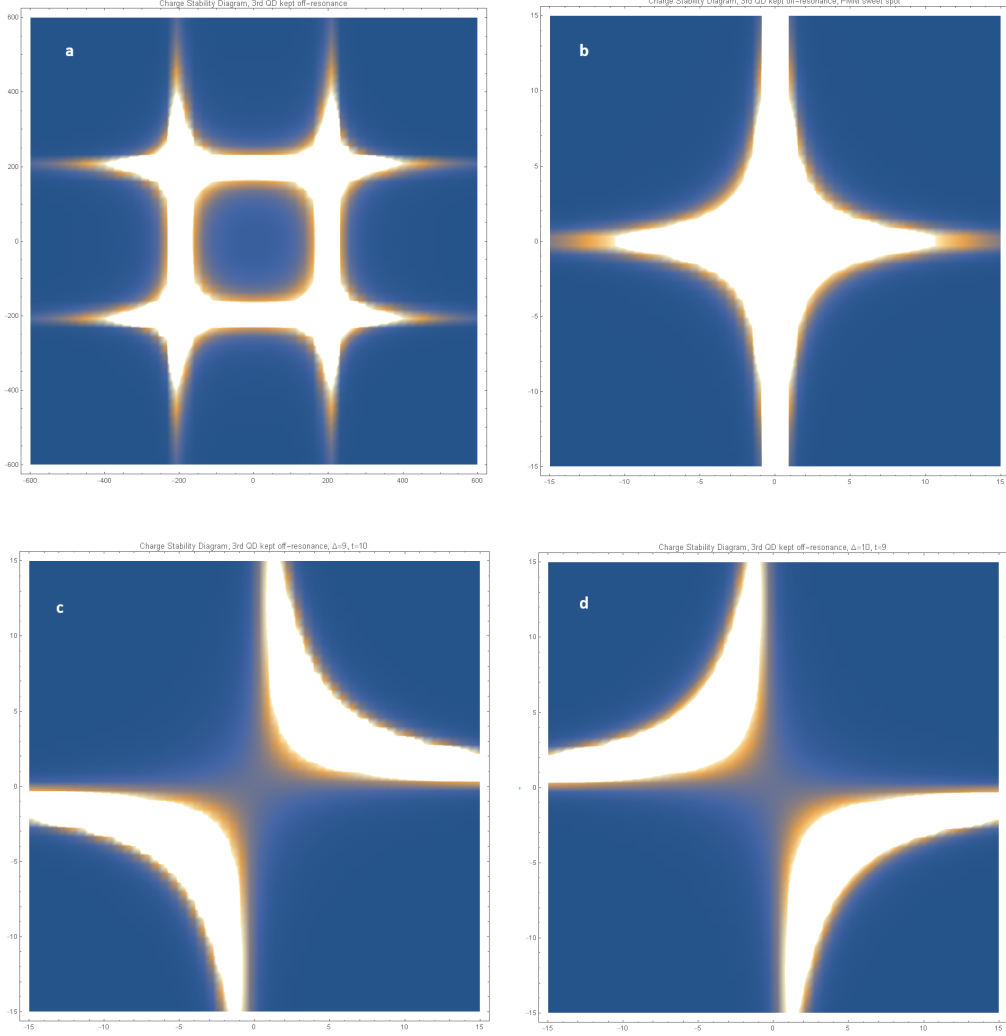


Figure 11: **Not to scale.** Depiction of the charge stability diagrams when the third quantum dot is kept off resonance. **a.** Whole charge stability diagram at the PMM sweet spot. Nine regions are clearly visible, corresponding to the occupation number states $|n_1, n_2, 0\rangle$, $n_1, n_2 \in \{0, 1, 2\}$. **b.** Isolated PMM state shifted at $(0,0)$. **c.** Gap opening in the PMM phase for $\Delta = 9, t = 10$, meaning ECT processes are dominating. **d.** Gap opening in the PMM phase for $\Delta = 10, t = 9$, meaning CAR are dominating. Note how the convexity of the gap flips in the two phases.

Three site Kitaev chain

The most general realization of a Kitaev chain, provided the setup in Fig.1, is the three site Kitaev chain as in eq.36. It should be noted now that the hopping and superconductive pairing terms are spread over two junctions rather than one, and a phase difference between the SC pairings will be acquired, that is: $\Delta_2 = e^{i\phi}|\Delta_2|$. Writing the hamiltonian explicitly:

$$H = \sum_{n=1}^3 \mu_n c_n^\dagger c_n + (t_1 c_1^\dagger c_2 + t_2 c_2^\dagger c_3 + |\Delta_1| c_1^\dagger c_2^\dagger + e^{i\phi} |\Delta_2| c_2^\dagger c_3^\dagger + h.c.) \quad (38)$$

Again the scattering matrix S will be derived by Weidenmuller's formula 25, but, since none of the QDs is kept off-resonance, and the system is depicted as in Fig. 4, the diagonal matrix W will have coupling terms to both leads:

$$W = \text{diag}(\sqrt{\Gamma_L}, 0, \sqrt{\Gamma_R}, -\sqrt{\Gamma_L}, 0, -\sqrt{\Gamma_R}) \quad (39)$$

And the conductance matrix will follow eq.34, this time being a 3x3 matrix. In our derivation of the conductance, we assume the same parameters as before, with a right lead coupling strength $\Gamma_R = 0.3\mu eV$.

It should be noted that the voltage difference of the left and right leads cannot be set to zero precisely: it is estimated [1] $0.1\mu V \leq \delta V \leq 1\mu V$, which means the phase difference will precess according to the Josephson equation: $\dot{\phi} = \frac{2e}{\hbar}\delta V$. Since over a period T_ϕ the phase will make a 2π jump, we get:

$$\frac{2\pi}{T_\phi} = \frac{2e}{\hbar}\delta V \quad (40)$$

From which $2ns \leq T_\phi \leq 20ns$. Since the lifetime of an electron in the QD is of the order $\tau_e \simeq \frac{\hbar}{\Gamma} \approx 1 ns \ll T_\phi$, the phase difference will be fixed over its lifetime. By measuring over long timescales ($\approx 1s$), it is safe to assume that the measured conductance will be averaged over the whole phase difference ensemble:

$$\langle G_{ij}^{(T)}(\omega) \rangle = \int_0^{2\pi} d\phi \frac{G_{ij}^{(T)}(\omega, \phi)}{2\pi} \quad (41)$$

The distribution of phase differences is taken to be uniform over $[0, 2\pi]$. Furthermore, we plotted (see Fig.13) the effect of the phase difference over the conductance G_{LL}, G_{RR} of the left and right leads, respectively. We can clearly notice a dip in the conductance at $\phi = \pi$. Interestingly, for small deviations from $\phi = \pi$, the conductance has a pronounced peak in both cases.

In Fig.12 we plotted the averaged conductance over the left lead, $G_{LL}(\omega)$, as a function of the detuning of every QD (first, second and third column). We notice zero-bias conductance peaks manifesting on both ends of the device, remaining stable against the detuning of any constituent QD.

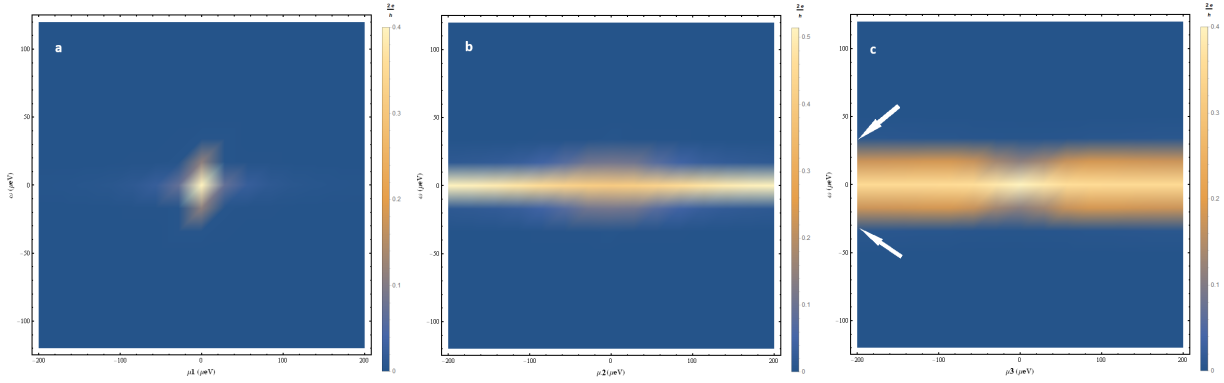


Figure 12: Numerical simulations of the conductance from the left lead, as a function of the detuning μ_i , $i \in \{1, 2, 3\}$ of quantum dots QD1 (a), QD2 (b) and QD3 (c). We observe zero bias conductance peaks (a., c.). By looking at the excited states when QD3 is off-resonance (white arrows in c., we can estimate the left couplings to be $2t_1 \approx 2\Delta_1 \approx 20\mu eV$.

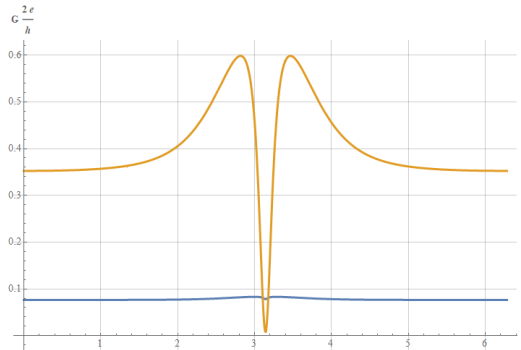


Figure 13: G_{LL} conductance from the left lead at resonance condition ($\mu_1 = \mu_2 = \mu_3 = 0$) vs phase difference ϕ in the range $[0, 2\pi]$. Clearly the conductance is periodic with period 2π with respect to ϕ .

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